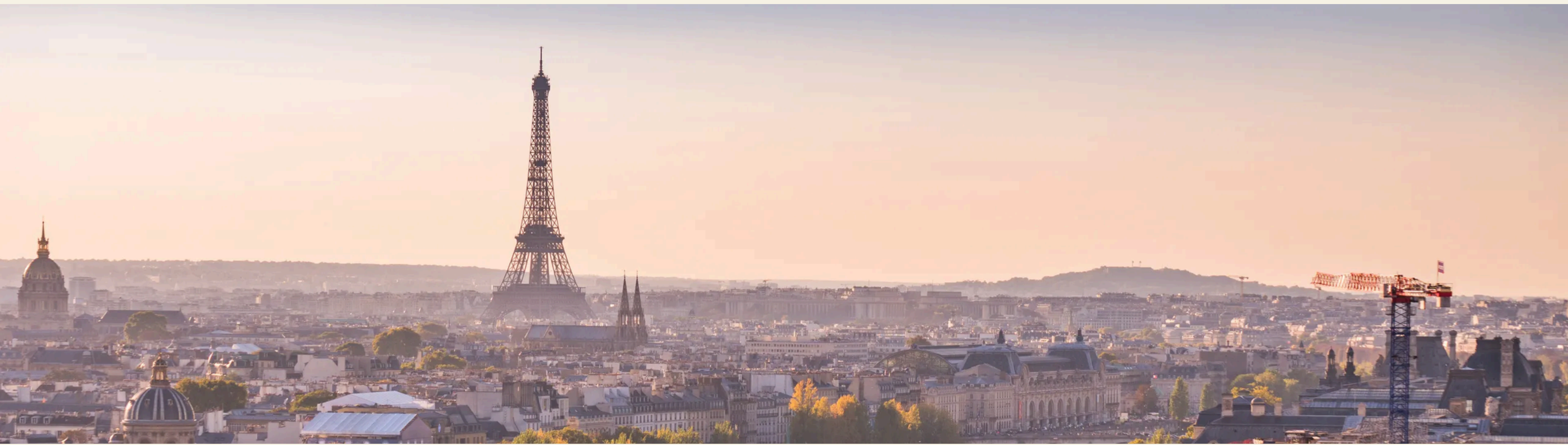


TESTING PLAN

Filip Łapiński
Adam Oporski
Gabriela Jastrzębska

UNIT TESTING

- **Purpose:** validating individual components to ensure they work as expected
- **Scope:** backend API endpoints, route optimization algorithms, frontend components (buttons, forms)
- **Backend:** Jest (Node.js), PyTest (Flask)
- **Frontend:** React Testing Library



INTEGRATION TESTING

- **Purpose:** verifying that different software modules interact correctly (e.g., frontend communicating with the backend, API handling database queries)
- **Scope:** route creation with multiple waypoints, user authentication and authorization flows, payment gateway interaction (premium subscriptions)
- **Tools/Frameworks:** Postman (API testing), Cypress (frontend-backend interactions)



USER ACCEPTANCE TESTING

- **Purpose:** ensuring the system meets user requirements and is ready for release
- **Scope:** full end-to-end testing of user journeys - creating, saving, and sharing routes; real-world simulation with test users evaluating functionality, usability, and performance
- **Tools/Frameworks:** manual testing, BrowserStack



Test Case	Description	Type	Expected Output	Tools
User registration & login	Validate that users can register, log in, and log out successfully.	Unit / Integration	Successful registration, login redirects to dashboard.	Jest, Postman
Route Creation	Test adding a new route with multiple waypoints.	Integration	Route saved in the database; visible in user profile.	Cypress, Postman
Collaborative Editing	Test if multiple users can edit the same route in real-time.	Integration	Changes reflected instantly for all users.	Cypress
Offline Access	Verify route download feature works, and routes can be accessed offline.	Integration / UAT	Download successful; route viewable without an internet connection.	Manual Testing
Payment Processing (Premium Feature)	Test premium subscription payment and activation process.	Integration	Payment processed; premium features unlocked.	Postman
Mobile Responsiveness	Validate UI across mobile, tablet, and desktop devices.	UAT	UI adjusts dynamically without layout issues.	BrowserStack, Cypress
Route Recommendations	Test the route suggestion algorithm based on user preferences.	Unit / Integration	Personalized route recommendations provided.	PyTest, Jest

TESTING TOOLS:

- Jest (Node.js) - a JavaScript testing framework primarily used for testing Node.js applications and frontend code; fast, easy to set up, supports snapshot testing, making it great for testing APIs, functions, and React components.
- PyTest (Flask) - a Python testing framework used to test Flask (or any Python) applications; simple, scalable, supports fixtures, parameterized testing, and plugins
- React Testing Library - a lightweight testing library for React applications; focuses on testing components in a way that mimics real user interactions, ensuring that the UI behaves correctly



TESTING TOOLS:

- Postman - an API testing tool that allows developers to manually or automatically test RESTful APIs; supports automated scripts and collections to streamline testing workflows
- Cypress - an end-to-end testing framework for web applications; operates directly in the browser, providing real-time feedback and a powerful debugging experience
- BrowserStack - a cloud-based platform that enables cross-browser and cross-device testing without requiring physical devices; offers real-time testing on different operating systems and browsers



TESTING SCHEDULE:

Phase	Test Type	Duration	Who Tests	Goal
Sprint 1-2	Unit Testing (Backend)	1 week	Developers	Ensure API endpoints work correctly.
Sprint 3	Unit Testing (Frontend)	1 week	Frontend Developers	Validate UI components and form inputs.
Sprint 4	Integration Testing	2 weeks	QA Engineers	Ensure frontend-backend communication.
Sprint 5	Performance Testing	1 week	QA Engineers	Test route load times and API response rates.
Sprint 6	UAT (Real Users)	2 weeks	Test Group (10 Users)	Validate end-to-end functionality.
Post-launch	Regression Testing	Ongoing (weekly)	QA Engineers	Test after every update to ensure nothing breaks.

THANK YOU
FOR YOUR
ATTENTION

Filip Łapiński
Adam Oporski
Gabriela Jastrzębska